



BobSim

TransientEval Lateral Transient Response

Step response and sustained-sine FRF

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Run Notes

- Representative step steer response
- Continuous sinusoidal responses
- Rise time uses 50 percent input to 90 percent response thresholds
- Frequency delays are reported at 0.5 Hz and 1.0 Hz
- Gain and phase slopes summarize the FRF trend
- Steering shown in degrees for readability

TransientEval Metrics Summary 15.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.05	s
Sideslip Rise Time (50-90%)	0.18	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.10	s
a_y DC Gain	31.85	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	0.039	rad/rad
Yaw DC Gain	2.10	(rad/s)/rad
Roll DC Gain	0.050	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	3.18	m/s^2
a_y Steady-State	2.68	m/s^2
a_y Overshoot	18.5	%
Yaw Steady-State	0.180	rad/s
Yaw Overshoot	0.030	rad/s
Roll Steady-State	0.004	rad
Roll Overshoot	0.001	rad
Settling Time	1.29	s

TransientEval Metrics Summary 15.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

DC Gain (a_y)	31.85	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.10	(rad/s)/rad
Peak Gain (a_y)	32.99	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.99	dB/dec
Gain Slope (r)	0.31	dB/dec
Phase Slope (a_y)	6.82	deg/dec
Phase Slope (r)	14.96	deg/dec

DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.007	s
Yaw/Steer Delay @ 0.5 Hz	0.025	s
Difference @ 0.5 Hz	-0.017	s
Ay/Yaw Delay @ 1.0 Hz	0.010	s
Yaw/Steer Delay @ 1.0 Hz	0.025	s
Difference @ 1.0 Hz	-0.015	s
Gain Variation	3.5	%
Fit Error (a_y)	8.57e-03	-
Fit Error (r)	6.35e-03	-

TransientEval Metrics Summary 20.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.72	s
Sideslip Rise Time (50-90%)	1.46	s
Yaw Rise Time (50-90%)	0.07	s
Yaw Peak Response Time	1.65	s
a_y DC Gain	55.06	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	-0.013	rad/rad
Yaw DC Gain	2.76	(rad/s)/rad
Roll DC Gain	0.088	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	4.59	m/s^2
a_y Steady-State	4.59	m/s^2
a_y Overshoot	0.1	%
Yaw Steady-State	0.232	rad/s
Yaw Overshoot	0.000	rad/s
Roll Steady-State	0.008	rad
Roll Overshoot	0.000	rad
Settling Time	2.07	s

TransientEval Metrics Summary 20.0 m/s

Frequency Domain - Sustained Sine

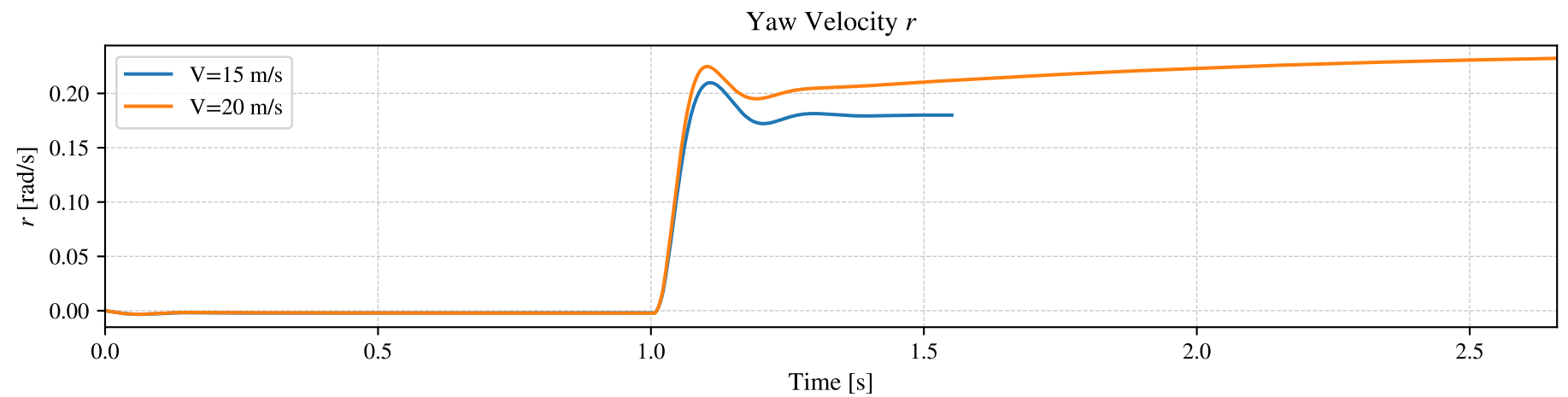
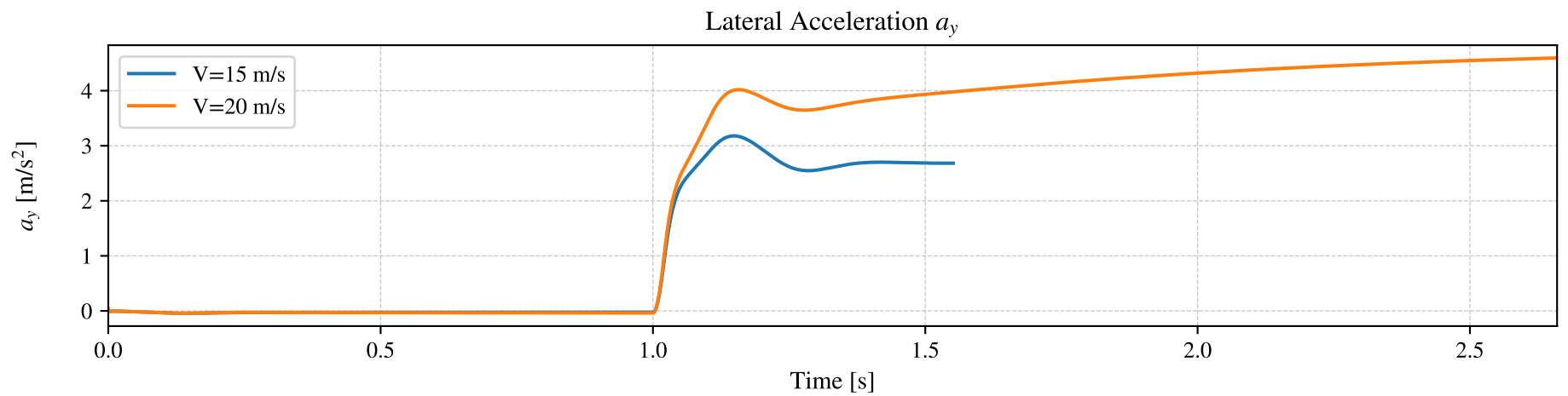
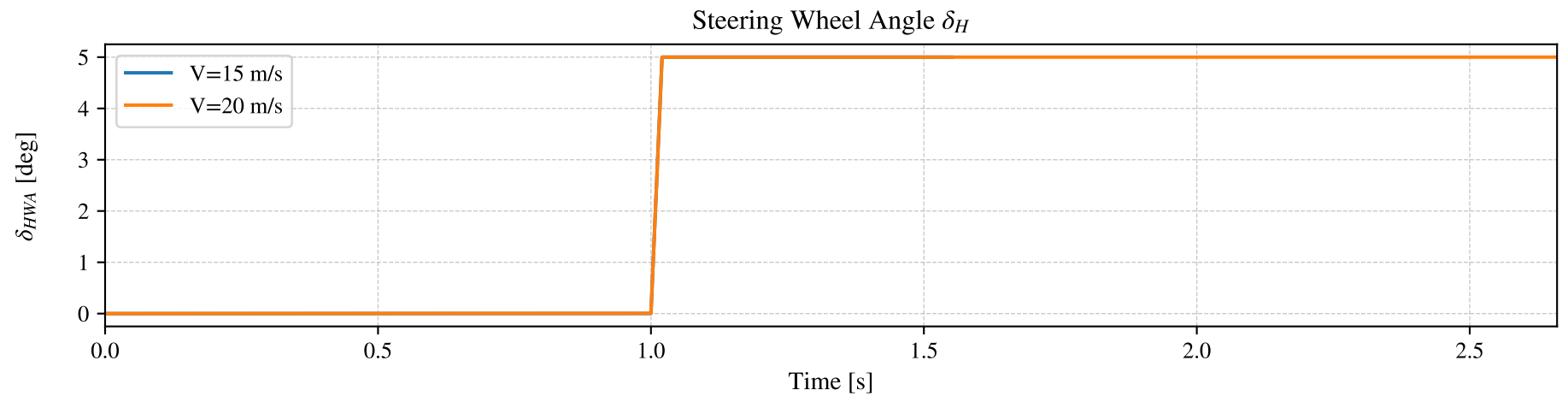
FREQUENCY RESPONSE - CORE

DC Gain (a_y)	55.06	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.76	(rad/s)/rad
Peak Gain (a_y)	55.06	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	0.50	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	-0.58	dB/dec
Gain Slope (r)	-0.27	dB/dec
Phase Slope (a_y)	21.38	deg/dec
Phase Slope (r)	17.54	deg/dec

DELAY / COUPLING

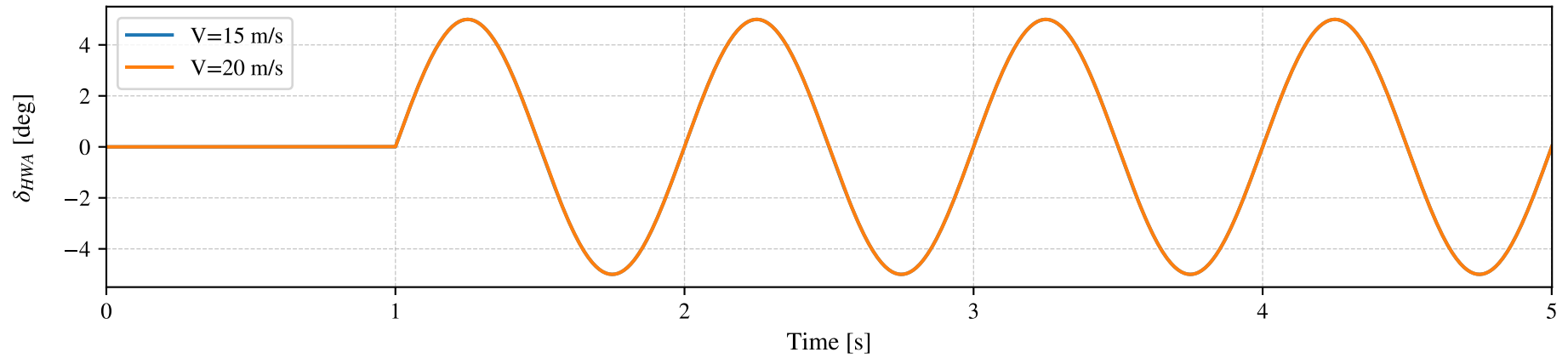
Ay/Yaw Delay @ 0.5 Hz	0.036	s
Yaw/Steer Delay @ 0.5 Hz	0.030	s
Difference @ 0.5 Hz	0.007	s
Ay/Yaw Delay @ 1.0 Hz	0.036	s
Yaw/Steer Delay @ 1.0 Hz	0.030	s
Difference @ 1.0 Hz	0.007	s
Gain Variation	2.1	%
Fit Error (a_y)	1.81e-02	-
Fit Error (r)	1.17e-02	-

Step Steer Response (TransientEval)

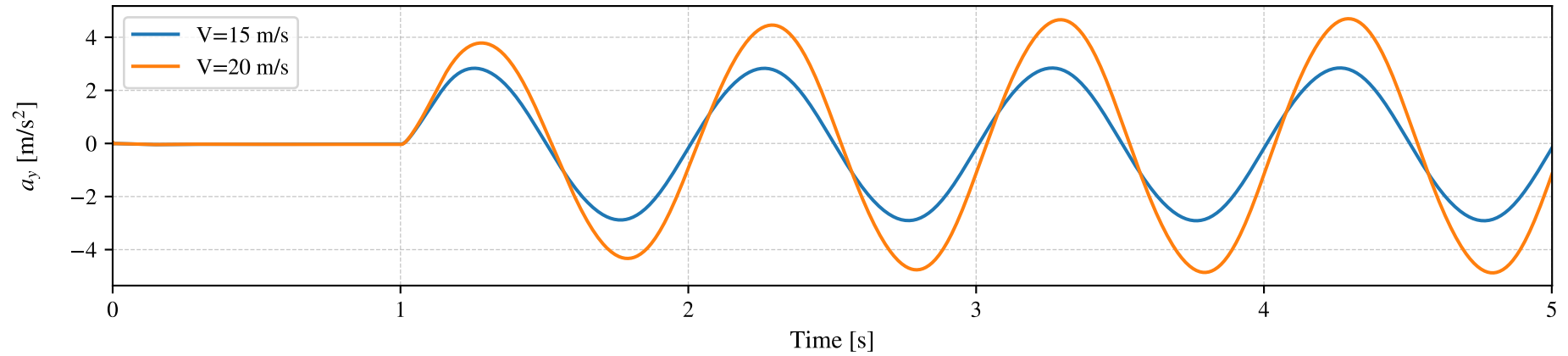


Continuous Sinusoidal Response

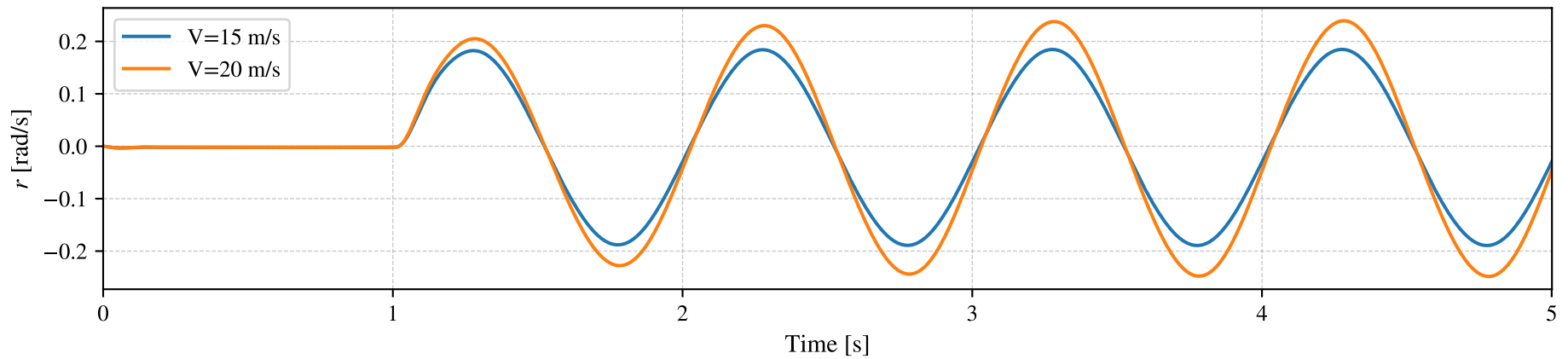
Steering Wheel Angle δ_H



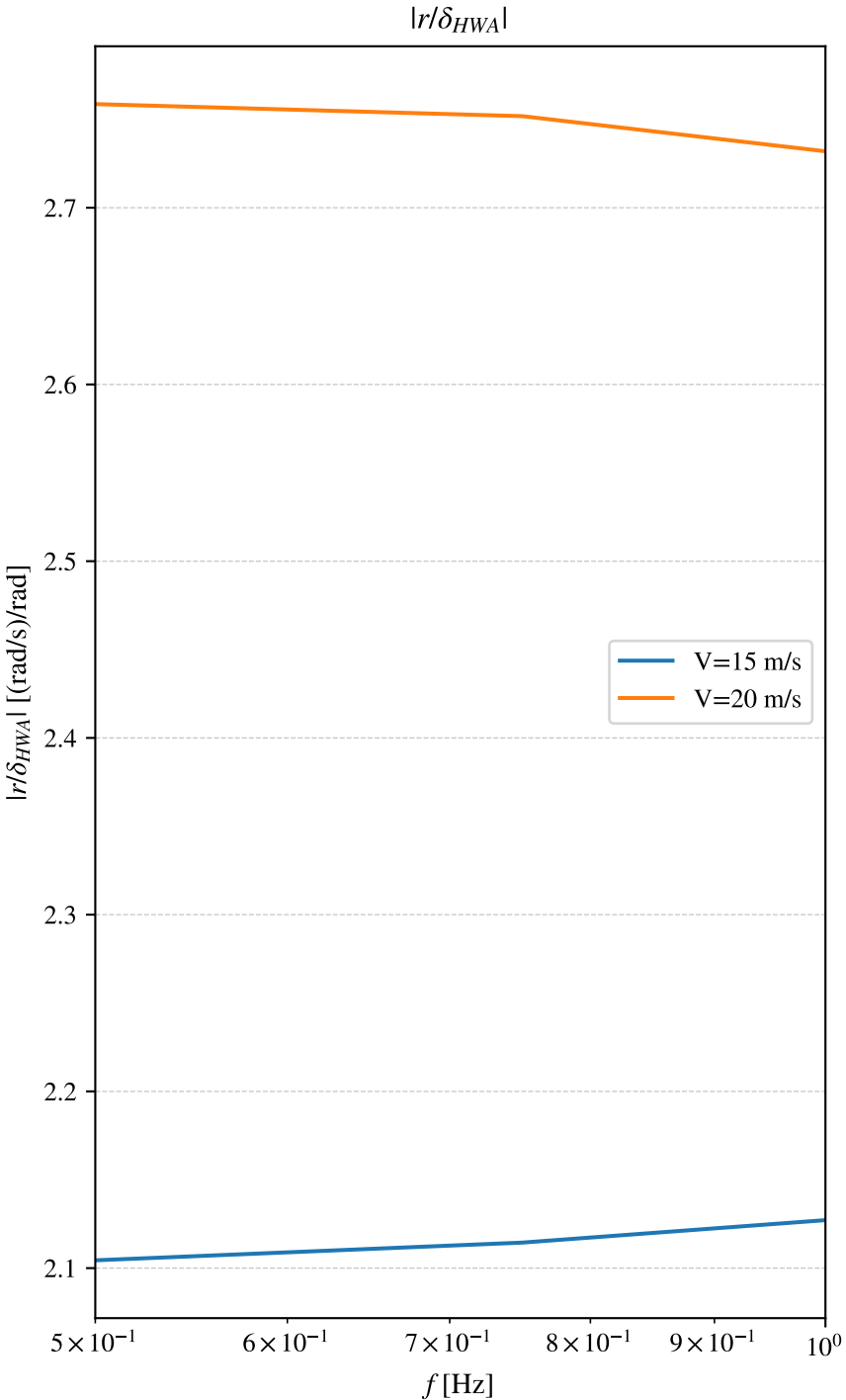
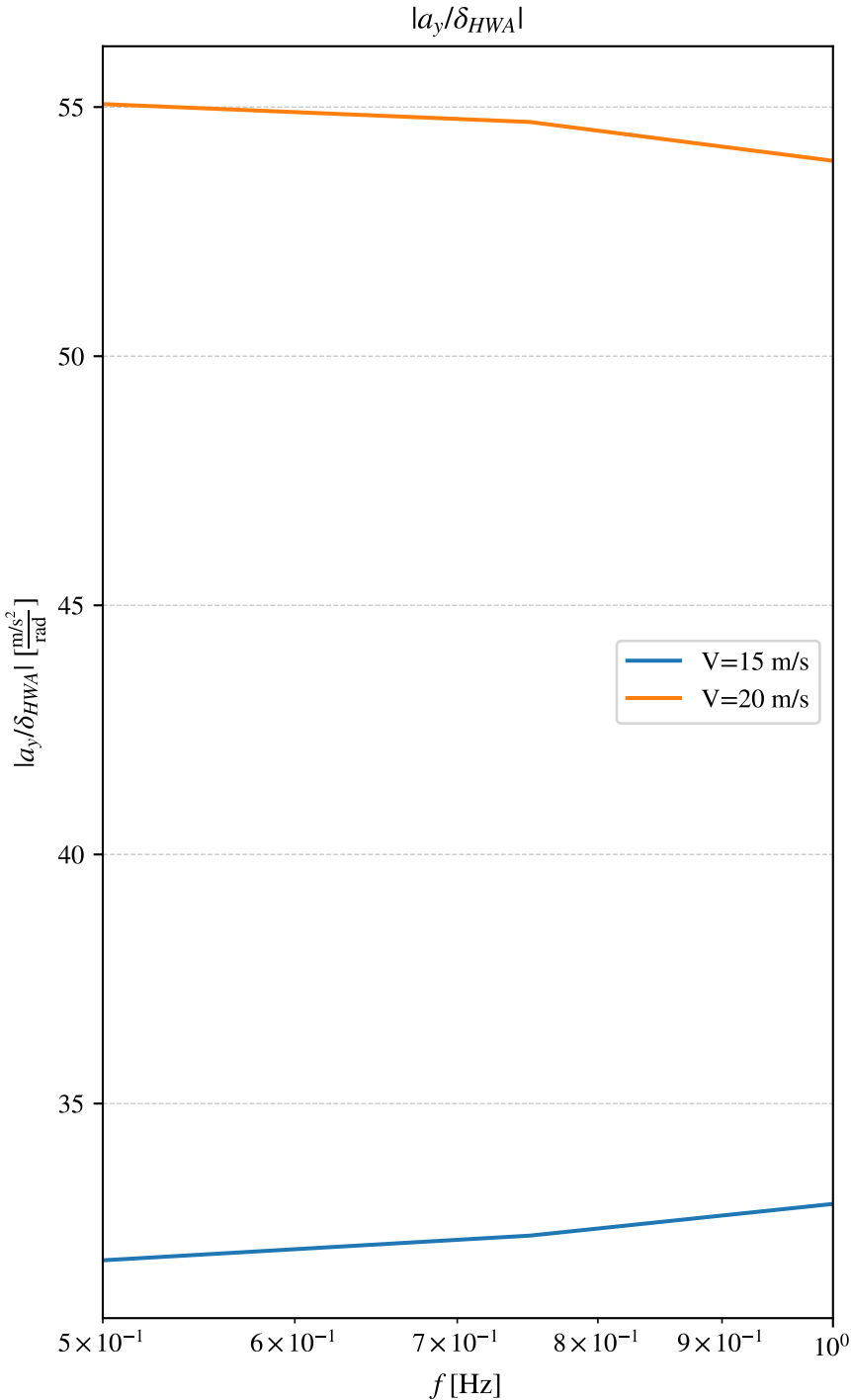
Lateral Acceleration a_y



Yaw Velocity r

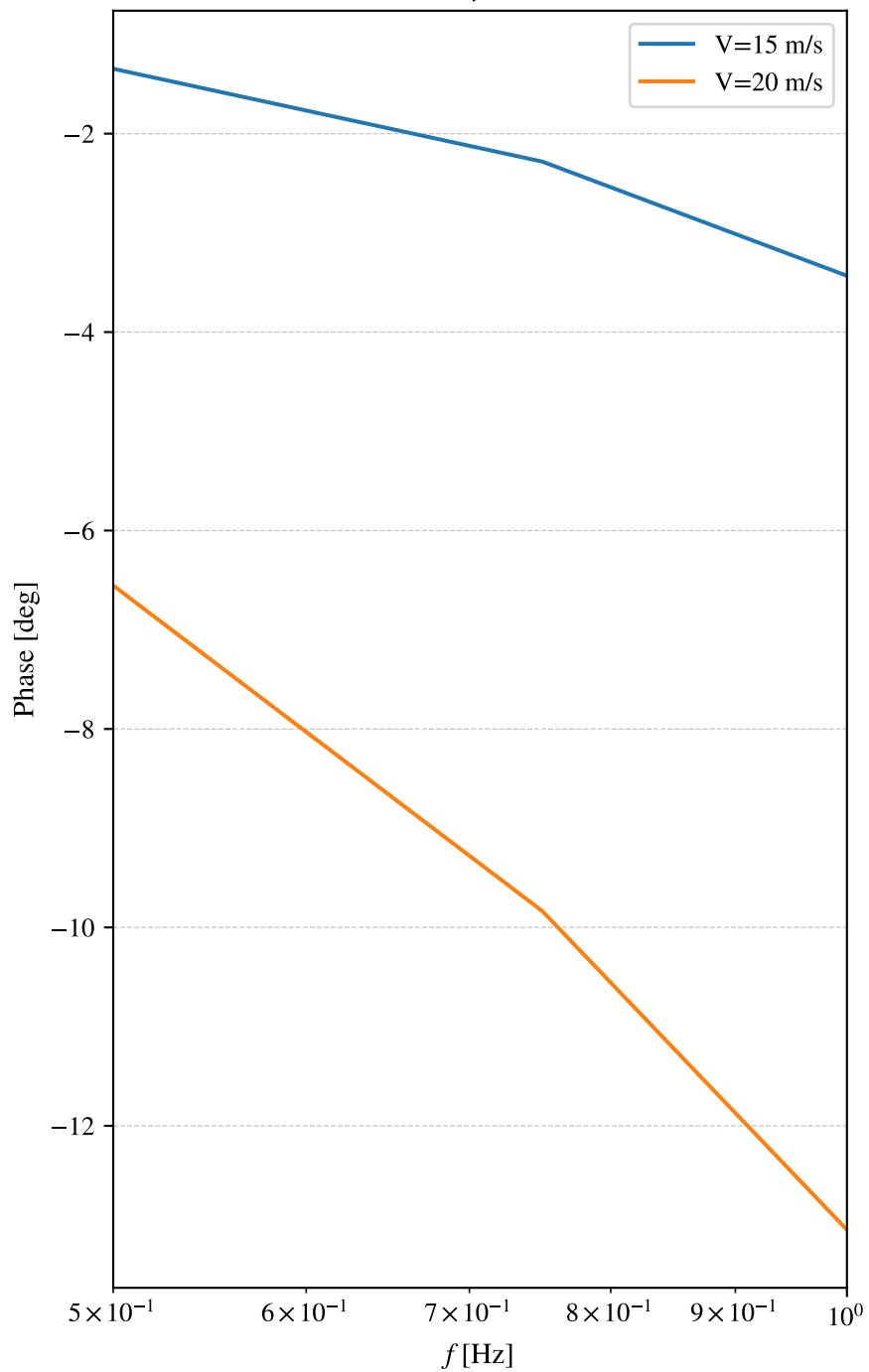


Frequency Response — Gain (TransientEval)



Frequency Response — Phase (TransientEval)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

